

SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Visualization Design Activity

Course Code:	DSA0502	Course Title:	Query Processing for Data Science
CO Assessed:	CO5	Assessment:	Tool 3– Visualization Design Activity
Weightage:	35%	Total Marks:	

CO5 Statement: Create meaningful visualizations using Pandas and Matplotlib, including charts, distributions, relationships, and interactive visual representations for effective communication

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Section Batch: /	D/2024	Date of Submission:	20/082026
Faculty In-charge:	Dr. B.Shyamala Gowri	Project Mode:	Individual Submission

Code of Conduct

I BHARANI DHARAN S (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: BHARANI DHARAN S

8	Frequency Visualization	Create a categorical dataset such as department-wise employee count. Represent the frequencies using a bar chart.	Frequency chart	10 min
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"""

Project 8: Counts and Frequencies

Topic : Department-wise Employee Count (Categorical Dataset)

Task : Represent the frequencies using a Bar Chart

Duration: 10 min

This script:

1. Loads / creates a categorical dataset (employee department data).
2. Computes the frequency (count) of employees in each department.
3. Displays the frequency table.
4. Plots a Bar Chart to visualize the frequencies.
5. Saves the chart as an image in the 'output' folder.

"""

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import os
```

```
# -----
```

```
# STEP 1: Load the categorical dataset
```

```
# -----
```

```
DATA_FILE = "employee_data.csv"
```

```
OUTPUT_DIR = "output"
```

```
os.makedirs(OUTPUT_DIR, exist_ok=True)
```

```
df = pd.read_csv(DATA_FILE)
```

```
print("----- Sample of Dataset -----")
```

```
print(df.head(), "\n")
```

```
# -----
```

```
# STEP 2: Calculate frequency (count) of each department
```

```
# -----
```

```
dept_counts = df["Department"].value_counts().sort_index()
```

```

print("----- Department-wise Employee Frequency -----")
print(dept_counts, "\n")

# Save the frequency table as CSV too
freq_table = dept_counts.reset_index()
freq_table.columns = ["Department", "EmployeeCount"]
freq_table.to_csv(os.path.join(OUTPUT_DIR, "department_frequency_table.csv"),
index=False)

# -----
# STEP 3: Plot Bar Chart of Frequencies
# -----
plt.figure(figsize=(8, 5))
bars = plt.bar(dept_counts.index, dept_counts.values, color="#4C72B0",
edgecolor="black")

# Add count labels on top of each bar
for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width() / 2, height + 0.1,
str(int(height)), ha="center", va="bottom", fontweight="bold")

plt.title("Department-wise Employee Count (Frequency Chart)", fontsize=13,
fontweight="bold")
plt.xlabel("Department")
plt.ylabel("Number of Employees")
plt.xticks(rotation=0)
plt.grid(axis="y", linestyle="--", alpha=0.6)
plt.tight_layout()

# -----
# STEP 4: Save the chart
# -----
output_path = os.path.join(OUTPUT_DIR, "department_frequency_bar_chart.png")
plt.savefig(output_path, dpi=200)
print(f'Bar chart saved to: {output_path}')

plt.show()

```

OUTPUT :

